Exact Worst-Case Coverage of the Jackknife+ and a 3/2 Law for Two-Fold Cross-Validation Conformal Prediction

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Abstract

The jackknife+, CV+, and cross-conformal predictors build distribution-free prediction intervals without sacrificing data to a calibration split, but their assumption-free guarantees are essentially coverage $1-2\alpha$ instead of $1-\alpha$. Barber et al. [2] asked whether the guarantee for K-fold CV+ can be sharpened. We determine the worst case in the two extreme regimes. For the jackknife+, the worst-case miscoverage over all distributions and all symmetric algorithms equals $(2\lfloor\alpha(n+1)\rfloor-1)_+/(n+1)$ for every n and every $\alpha \le 1/2$. The matching construction uses i.i.d. data and an explicit algorithm built from a "zigzag" permutation that realizes a regular tournament through leave-two-out fits. For two-fold CV+ and cross-conformal prediction the factor 2 is not sharp: the worst-case miscoverage converges to $\frac{3}{2}\alpha$ as the fold size grows, with explicit non-asymptotic bounds. The reason is that the two fold p-values are asymptotically conditionally i.i.d. given the test point, and the average of two conditionally i.i.d. p-values needs only the factor 3/2. The same sharp inequality gives an exactly valid rule for averaging two independent random splits of split conformal prediction. For general K we give finite-sample bounds that improve both bounds of Barber et al. [2]. We also show that for every fixed K the large-fold worst case is at least $\alpha\, g_K(\alpha)$, where g_K is the sharp averaging constant of K conditionally i.i.d. p-values. Since $g_K \to 2$, the factor 2 returns as the number of folds grows. The exact worst case for fixed $K \ge 3$ remains open. All results are proved, and exact computation is used only to check them.

1 Introduction

Conformal prediction turns any regression algorithm into a prediction interval with a finite-sample coverage guarantee that holds for every data distribution [1, 13, 20]. Split conformal prediction achieves coverage $1-\alpha$ exactly but sets aside part of the data for calibration. Methods based on cross-validation avoid this loss: Vovk's cross-conformal predictor [18], and the jackknife+ and CV+ of Barber et al. [2], reuse every observation for both fitting and calibration. The price is a weaker guarantee. Barber et al. [2] proved that the jackknife+ covers with probability at least $1-2\alpha$ for every distribution and every symmetric algorithm, and that K-fold CV+ covers with probability at least

$$1-2\alpha-\min\left\{\tfrac{2(1-1/K)}{n/K+1},\ \tfrac{1-K/n}{K+1}\right\} \ \ge\ 1-2\alpha-\sqrt{2/n}. \tag{1}$$

For cross-conformal prediction, and hence for CV+ (Section 2), Gasparin and Ramdas [7] state the slightly sharper first term $2(1-\alpha)(1-1/K)/(n/K+1)$. Barber et al. [2] also constructed a distribution and an unstable algorithm for which the jackknife+ covers with probability at most $1-2\alpha+6\sqrt{\log n/n}$, so the factor 2 cannot be removed for the jackknife+ in general, and they wrote that it would be interesting to see whether the guarantee for K-fold CV+ can be sharpened.

This paper studies the *worst-case miscoverage*

$$\mathfrak{W}(\text{method}; n, \alpha) \ =\ \sup_{P,\mathcal{A}}\ \mathbb{P}\big(Y_{n+1} \notin \widehat{C}_{n,\alpha}(X_{n+1})\big),$$

the supremum over all data distributions P and all symmetric regression algorithms $\mathcal{A}$ (several of our upper bounds hold without symmetry), and asks for its exact value. We answer this in the two extreme regimes of the fold structure and improve the known bounds in between.

Contributions.

1. **The jackknife+ is exactly characterized** (Theorem 2). For every $n \geq 2$ and $\alpha \in (0, 1/2]$,

$$\mathfrak{W}(\text{jackknife+}; n, \alpha) = \frac{\big(2\lfloor \alpha(n+1) \rfloor - 1\big)_+}{n+1},$$

   and the same holds for leave-one-out cross-conformal prediction. The upper bound is an integer refinement of the tournament argument of Barber et al. [2]. The new part is the matching lower bound for i.i.d. data at every n. It is not automatic: a regression algorithm fitted without two points cannot see which two points were removed, so not every comparison pattern (tournament) between residuals can be realized. We give an explicit algorithm based on a "zigzag" permutation that realizes a regular tournament on the $2\lfloor \alpha(n+1) \rfloor - 1$ largest responses (Lemma 3). The zigzag rule decides comparisons between distinct positions by a fixed ranking of positions, and we show that no rule of this kind realizes the familiar rotational tournament (Proposition 5).

2. **For two folds the factor is** $3/2$**, not** 2 (Theorem 10). For two-fold CV+ and two-fold cross-conformal prediction with folds of size m,

$$\lim_{m\to\infty} \mathfrak{W}(\text{2-fold CV+}; 2m, \alpha) = \lim_{m\to\infty} \mathfrak{W}(\text{2-fold cross-conformal}; 2m, \alpha) = \tfrac{3}{2}\alpha,$$

   with explicit rates $O(\sqrt{\log m/m})$ on both sides. The reason is structural: given the test point, the two fold p-values are asymptotically conditionally i.i.d., and for conditionally i.i.d. p-values the arithmetic mean needs the factor $3/2$ rather than the factor 2 that is sharp under arbitrary dependence (Lemma 8). The same lemma gives an exact finite-sample guarantee for aggregated split conformal prediction with two independent random splits (Corollary 9).

3. **Sharper finite-sample bounds for** K**-fold CV+** (Theorem 6). A discrete Markov-type averaging bound and an exact integer counting bound each improve the corresponding term of (1), the first also in its sharper form. Together they give coverage at least $1 - 2\alpha$, with no extra term, whenever $K = 2$ or $n/K \leq 2$. For $K \in \{5, 10, 20\}$ and $n \geq 1000$ they shrink the extra term of the best previous bound by a factor of at least 2.1 in all settings we examined (Section 8).

4. **Many folds bring the factor** 2 **back** (Section 6). For every fixed K,

$$\liminf_{m\to\infty} \mathfrak{W}(K\text{-fold CV+}; Km, \alpha) \;\geq\; \alpha\, g_K(\alpha),$$

   where g_K is the sharp averaging constant of K conditionally i.i.d. p-values (Proposition 12). The construction hashes the training sets so that the fold loss counts are exactly i.i.d. given the test point, for every fold size. For $K = 2$, $g_2(\alpha) = 3/2$ exactly for $\alpha \leq 1/2$ (Lemma 8). We prove $g_K(\alpha) \to 2$ as $K \to \infty$, so the large-fold worst case tends to 2α as the number of folds grows (Corollary 13). The exact value of g_K for $K \geq 3$ is unknown, and so is the exact worst case for fixed $K \geq 3$.

Every result above is proved. Computation plays a supporting role only (Section 7). It checks the constructions by exact enumeration and exact integration, compares them with direct simulation of the intervals, and supplies numerical lower bounds on g_K for $3 \leq K \leq 30$ (Table 2). These numbers are lower bounds from restricted families, not values of g_K.

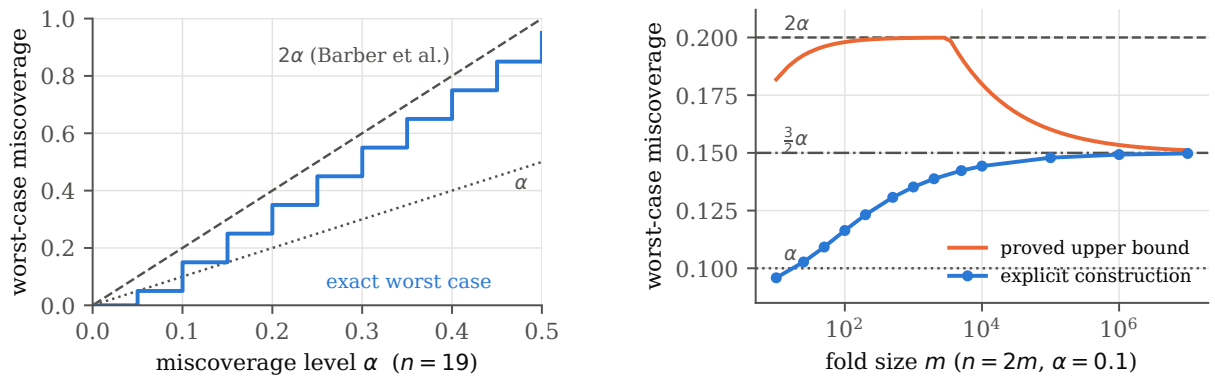


Figure 1: Left: exact worst-case miscoverage of the jackknife+ for $n = 19$ (Theorem 2); the known guarantee is the line 2α. Right: two-fold CV+ with $\alpha = 0.1$. The orange curve is the best proved upper bound (Theorems 6 and 10); the blue points are the exact miscoverage of our construction (Section 5), optimized over β; both converge to $\frac{3}{2}\alpha$.

Related work. Cross-conformal prediction was introduced by Vovk [18], who noted that no theoretical validity results were available and gave leave-one-out examples, under exchangeable (permutation) distributions, whose error rate exceeds the nominal level. Vovk et al. [21] extended the construction to predictive distributions and recommended doubling the p-values to restore validity. Barber et al. [2] proved the $1 - 2\alpha$ guarantees for the jackknife+ and CV+, related CV+ to cross-conformal prediction, and showed that the factor 2 is necessary for the jackknife+ asymptotically. Their proofs count "strange" points in a comparison tournament, an argument in the spirit of Landau's theorem on score sequences [12]; the same device is used for the jackknife+-after-bootstrap [11] and in the treatment of Angelopoulos et al. [1]. Under algorithmic stability the jackknife+ covers at nearly the nominal level [2], and stability also yields training-conditional guarantees [3, 14]; our results concern the assumption-free regime. Gasparin and Ramdas [7] gave variants of cross-conformal prediction with guaranteed coverage $1-2\alpha$ and smaller sets. Aggregated conformal predictors, which average p-values over many random splits, were introduced by Carlsson et al. [4]; Linusson et al. [15] describe their validity as not fully understood.

On the probability side, Vovk and Wang [19] showed that twice the arithmetic mean of arbitrarily dependent p-values is a valid p-value and that the factor 2 cannot be improved. Motivated by de Finetti's theorem, Choi and Kim [5] asked whether exchangeability helps. They showed that the factor 2 remains sharp for exchangeable p-values, and that it can be reduced under a specific location-mixture model with unimodality and symmetry conditions. Gasparin et al. [8] obtained strict improvements for exchangeable p-values through non-symmetric and randomized rules. P-values that are i.i.d. conditionally on the data arise whenever a randomized procedure, such as a random data split, is repeated. For such p-values, Mattner [16] determined the sharp calibration of each single order statistic and asked whether competitive combinations exist that are not functions of one order statistic. DiCiccio et al. [6] construct exact tests from many data splits, including tests based on the mean and the median of the split p-values, and Guo and Shah [9] develop rank-transformed subsampling for this setting. As the number of splits grows, the mean tends to the conditional expectation of a p-value given the data. Twice such a conditional expectation is a valid p-value [17], and the factor 2 cannot be improved [22]. Our Lemma 8 shows that for two conditionally i.i.d. p-values (mixtures of i.i.d. laws) the arithmetic mean needs only the factor $3/2$,

and that this is sharp. We found no prior statement of this constant, of the exact finite-sample worst case of the jackknife+ for i.i.d. data, or of the 3/2 law for two-fold cross-validation conformal methods.

2 Setup

Let $Z_i = (X_i, Y_i) \in \mathcal{X} \times \mathbb{R}$, $i = 1, \dots, n+1$, be i.i.d. from a distribution P; the first n points are training data and Z_{n+1} is the test point. A regression algorithm $\mathcal{A}$ maps a finite collection of data points to a function $\mathcal{X} \to \mathbb{R}$. It is *symmetric* if its output does not depend on the order of its inputs. Throughout, $\alpha \in (0,1)$ and

$$\ell := \lfloor \alpha(n+1) \rfloor, \qquad t := \lceil (1-\alpha)(n+1) \rceil = n+1-\ell, \qquad L := n - t = \ell - 1.$$

For reals $v_1, \dots, v_n$, $\widehat{q}^{+}_{n,\alpha}\{v_i\}$ is the t-th smallest value ($+\infty$ if $t > n$) and $\widehat{q}^{-}_{n,\alpha}\{v_i\}$ is the ℓ-th smallest value ($-\infty$ if $\ell = 0$).

Jackknife+. Let $\widehat{\mu}_{-i} = \mathcal{A}\big((Z_j)_{j \in [n] \setminus \{i\}}\big)$ and $R_i = |Y_i - \widehat{\mu}_{-i}(X_i)|$. The jackknife+ interval is

$$\widehat{C}^{\mathrm{J+}}_{n,\alpha}(x) = \Big[\widehat{q}^{-}_{n,\alpha}\{\widehat{\mu}_{-i}(x) - R_i\},\ \widehat{q}^{+}_{n,\alpha}\{\widehat{\mu}_{-i}(x) + R_i\}\Big].$$

CV+ and cross-conformal. Let $n = Km$ and split $[n]$ into folds $S_1, \dots, S_K$ of size m; write $k(i)$ for the fold of i. Let $\widehat{\mu}_{-S_k} = \mathcal{A}\big((Z_j)_{j \notin S_k}\big)$ and $R^{\mathrm{CV}}_i = |Y_i - \widehat{\mu}_{-S_{k(i)}}(X_i)|$. The CV+ interval is

$$\widehat{C}^{\mathrm{CV+}}_{n,K,\alpha}(x) = \Big[\widehat{q}^{-}_{n,\alpha}\{\widehat{\mu}_{-S_{k(i)}}(x) - R^{\mathrm{CV}}_i\},\ \widehat{q}^{+}_{n,\alpha}\{\widehat{\mu}_{-S_{k(i)}}(x) + R^{\mathrm{CV}}_i\}\Big],$$

and the (absolute-residual) cross-conformal set is

$$\widehat{C}^{\mathrm{CC}}_{n,K,\alpha}(x) = \Big\{y \in \mathbb{R} :\ \textstyle\sum_{i=1}^{n} \mathbf{1}\big\{R^{\mathrm{CV}}_i < |y - \widehat{\mu}_{-S_{k(i)}}(x)|\big\} < t\Big\}.$$

With $K = n$ these are the jackknife+ and leave-one-out cross-conformal prediction. The argument of Lemma 1 below, applied to an arbitrary y in place of Y_{n+1}, shows $\widehat{C}^{\,\mathrm{CC}} \subseteq \widehat{C}^{\,\mathrm{CV+}}$ (a containment also noted by Barber et al. [2]), so the cross-conformal set is never more conservative.

The loss counts. For the test point and fold k define

$$X_k := \#\big\{i \in S_k :\ R^{\mathrm{CV}}_i \ \geq\ |Y_{n+1} - \widehat{\mu}_{-S_k}(X_{n+1})|\big\} \in \{0, \dots, m\}, \tag{2}$$

the number of points of fold k whose residual is at least the test residual under the same fitted model. The next lemma is implicit in Barber et al. [2]; we record it because every result below starts from it.

Lemma 1. *For the cross-conformal set, $Y_{n+1} \notin \widehat{C}^{\mathrm{CC}}_{n,K,\alpha}(X_{n+1})$ if and only if $\sum_{k=1}^{K} X_k \leq L$. For CV+, the event $Y_{n+1} \notin \widehat{C}^{\mathrm{CV+}}_{n,K,\alpha}(X_{n+1})$ implies $\sum_k X_k \leq L$.*

Proof. Write $\mu_i = \widehat{\mu}_{-S_{k(i)}}(X_{n+1})$ and $E_i = Y_{n+1} - \mu_i$. By definition, $Y_{n+1} \notin \widehat{C}^{\,\mathrm{CC}}$ if and only if $\#\{i : R^{\mathrm{CV}}_i < |E_i|\} \geq t$. Equivalently $\#\{i : R^{\mathrm{CV}}_i \geq |E_i|\} \leq n - t = L$, which is $\sum_k X_k \leq L$. If $Y_{n+1} > \widehat{q}^{+}_{n,\alpha}\{\mu_i + R^{\mathrm{CV}}_i\}$, then at least t indices satisfy $R^{\mathrm{CV}}_i < E_i \leq |E_i|$. If $Y_{n+1} < \widehat{q}^{-}_{n,\alpha}\{\mu_i - R^{\mathrm{CV}}_i\}$, then fewer than ℓ of the values $\mu_i - R^{\mathrm{CV}}_i$ are at most Y_{n+1}, so at least $n - \ell + 1 = t$ indices satisfy $R^{\mathrm{CV}}_i < -E_i \leq |E_i|$. □

Every CV+ result below therefore also holds for cross-conformal prediction, and every upper bound is proved for the larger quantity $\mathbb{P}(\sum_k X_k \le L)$. We write $\mathfrak{W}_n^{\mathrm{J+}}(\alpha)$, $\mathfrak{W}_{n,K}^{\mathrm{CV+}}(\alpha)$ and $\mathfrak{W}_{n,K}^{\mathrm{CC}}(\alpha)$ for the worst-case miscoverages, with the supremum over all P and all symmetric $\mathcal{A}$. By Lemma 1, $\mathfrak{W}^{\mathrm{CV+}} \le \mathfrak{W}^{\mathrm{CC}}$. Symmetry is used only by the counting arguments: the averaging bound B_{avg} of Theorem 6 and the upper bound of Theorem 10(i) hold for arbitrary algorithms, and all our constructions use symmetric algorithms.

3 The jackknife+ is exactly characterized

Theorem 2. *For every $n \ge 2$ and $\alpha \in (0, 1/2]$,*

$$\mathfrak{W}_n^{\mathrm{J+}}(\alpha) \;=\; \mathfrak{W}_{n,n}^{\mathrm{CC}}(\alpha) \;=\; \frac{\max\{2\ell - 1,\, 0\}}{n+1}, \qquad \ell = \lfloor \alpha(n+1) \rfloor.$$

The supremum is attained by $X \sim \mathrm{Unif}[0,1]$, $Y = X$, and the explicit symmetric algorithm (5) below. Equivalently, for $\alpha \ge 1/(n+1)$ the exact worst-case coverage is $1 - 2\alpha + \frac{1+2\{\alpha(n+1)\}}{n+1}$, where $\{x\} = x - \lfloor x \rfloor$.

The factor 2 is thus attained up to at most $3/(n+1)$ for every n, and exactly $2\alpha - 1/(n+1)$ is attained whenever $\alpha(n+1)$ is an integer.

3.1 Upper bound

This is the argument of Barber et al. [2] with integer counting. Augment the fits: for $i \ne j$ in $[n+1]$ let $\widetilde{\mu}_{-(i,j)} = \mathcal{A}\big((Z_k)_{k \notin \{i,j\}}\big)$, $R_{ij} = |Y_i - \widetilde{\mu}_{-(i,j)}(X_i)|$ and $W_{ij} = \mathbf{1}\{R_{ij} > R_{ji}\}$. Since $\widehat{\mu}_{-i} = \widetilde{\mu}_{-(i,n+1)}$, Lemma 1 with $K = n$ shows that miscoverage implies $\sum_{i \le n} W_{n+1,i} \ge t$. Call j *strange* if $\sum_{i \ne j} W_{ji} \ge t$ and let S be the strange set. A strange point has at most $n - t = \ell - 1$ indices i with $W_{ji} = 0$. For distinct strange j, j' at least one of $W_{jj'}, W_{j'j}$ is 0. Counting these zeros,

$$\binom{|S|}{2} \le |S|(\ell - 1), \qquad \text{so} \qquad |S| \le 2\ell - 1 \;\; (\text{or } S = \emptyset).$$

The joint law of (R_{ij}) is invariant under permutations of $[n+1]$, because the data are exchangeable and $\mathcal{A}$ is symmetric. Hence $\mathbb{P}(n+1 \in S) = \mathbb{E}|S|/(n+1) \le (2\ell - 1)/(n+1)$. If $\ell = 0$ then $t = n + 1 > n$ and the interval is $\mathbb{R}$.

3.2 Lower bound: realizing a regular tournament

The upper bound is attained if and only if, with probability one, exactly $2\ell - 1$ points are strange and each of them beats every other point except exactly $\ell - 1$ of the strange ones, i.e. the strange points carry a *regular* tournament. The difficulty is that the comparison W_{ab} is decided by the single fit $\widetilde{\mu}_{-(a,b)}$, which sees neither Z_a nor Z_b. The fit evaluates each query point on its own, so all it can use about a query x is x itself and its position relative to the $n - 1$ points it was trained on. We now show that this is enough.

Positions. Order the $n+1$ responses decreasingly, $z_1 > z_2 > \cdots > z_{n+1}$ (ties have probability zero), and call the index the *rank*. For a training collection T and a query value u, let $p_T(u) = \#\{(x', y') \in T : \; x' > u\}$. For a pair of ranks $a < b$ the fit $\widetilde{\mu}_{-(a,b)}$ is trained on all points except these two, so

$$p(z_a) = a - 1, \qquad p(z_b) = b - 2. \tag{3}$$

The position of the upper point counts all $a-1$ points above it; the lower point misses the removed point z_a. This offset of one is what makes intransitive comparisons possible.

The zigzag permutation. Let $h = \ell - 1 \geq 1$. Index the $2h$ "gaps" between consecutive strange ranks by $c \in \{0, \ldots, 2h-1\}$ (gap c lies between ranks $c+1$ and $c+2$) and define $\gamma : \{0, \ldots, 2h-1\} \to \{0, \ldots, 2h-1\}$ and a direction $\tau_c \in \{\uparrow, \downarrow\}$ by

$$\gamma_c = \begin{cases} h-1-c, & c < h,\ c \text{ even}, \\ h+c, & c < h,\ c \text{ odd}, \\ c-h, & c \geq h,\ c \text{ even}, \\ 3h-1-c, & c \geq h,\ c \text{ odd}, \end{cases} \qquad \tau_c = \begin{cases} \uparrow, & (c < h,\ c \text{ even}) \text{ or } (c \geq h,\ c \text{ odd}), \\ \downarrow, & \text{otherwise.} \end{cases}$$

For example $h = 4$ gives $\gamma = (3, 5, 1, 7, 0, 6, 2, 4)$ and $\tau = (\uparrow, \downarrow, \uparrow, \downarrow, \downarrow, \uparrow, \downarrow, \uparrow)$. The four cases of γ are injective maps into $[0, h-1]$, $[h+1, 2h-1]$, $[0, h-1]$ and $[h, 2h-1]$, respectively, with values of parity $h-1$, $h-1$, h and h. These four sets of values are pairwise disjoint, so γ is injective and hence a permutation of $\{0, \ldots, 2h-1\}$.

Lemma 3 (zigzag tournament)**.** *On the vertex set $\{1, \ldots, 2h+1\}$ orient each pair $a < b$ as follows: if $b = a+1$, then a beats b if and only if $\tau_{a-1} = \uparrow$; if $b \geq a+2$, then a beats b if and only if $\gamma_{a-1} > \gamma_{b-2}$. This tournament is regular: every vertex beats exactly h others.*

Proof. For gap c let $A_c = \#\{c' < c : \gamma_{c'} < \gamma_c\}$ and $B_c = \#\{c' > c : \gamma_{c'} < \gamma_c\}$, so $A_c + B_c = \gamma_c$ because γ is a permutation of $\{0, \ldots, 2h-1\}$. Vertex v is compared through gap $v-2$ with every vertex above it and through gap $v-1$ with every vertex below it. It beats a non-adjacent vertex $a < v-1$ if and only if $\gamma_{a-1} < \gamma_{v-2}$, which gives A_{v-2} wins; it beats a non-adjacent vertex $b > v+1$ if and only if $\gamma_{v-1} > \gamma_{b-2}$, which gives B_{v-1} wins. Adding the two adjacent comparisons,

$$\text{wins}(v) = \underbrace{A_{v-2} + \mathbf{1}\{\tau_{v-2} = \downarrow\}}_{=: \lambda_{v-2}} + \underbrace{B_{v-1} + \mathbf{1}\{\tau_{v-1} = \uparrow\}}_{=: \rho_{v-1}}, \tag{4}$$

with $\lambda_{-1} = \rho_{2h} = 0$. Note $\lambda_c + \rho_c = \gamma_c + 1$. Let $Q_c = \sum_{c' \leq c} (\gamma_{c'} - (h-1))$. We claim that $A_c + \mathbf{1}\{\tau_c = \downarrow\} = Q_c$ for every c. Granting this, $\lambda_c = Q_c$ and $\rho_c = \gamma_c + 1 - Q_c$, so for $2 \leq v \leq 2h$, $\text{wins}(v) = Q_{v-2} + \gamma_{v-1} + 1 - Q_{v-1} = h$; also $\text{wins}(1) = \rho_0 = \gamma_0 + 1 - Q_0 = h$ and $\text{wins}(2h+1) = \lambda_{2h-1} = Q_{2h-1} = \binom{2h}{2} - 2h(h-1) = h$.

To verify the claim, use $A_c = \gamma_c - \#\{c' > c : \gamma_{c'} < \gamma_c\}$ and a direct computation of Q_c. In the first half, consecutive gaps $(2i, 2i+1)$ have $\gamma_{2i} + \gamma_{2i+1} = 2h$, so $Q_c = 0$ for even $c < h$ and $Q_c = c+1$ for odd $c < h$. For even $c < h$ every smaller value lies at a later gap, so $A_c = 0$. For odd $c < h$ the value $h + c$ exceeds all c earlier values, so $A_c = c$. In the second half a direct summation gives $Q_c = c - h + 1$ for even $c \geq h$ and $Q_c = h$ for odd $c \geq h$ (for both parities of h). For even $c \geq h$ the value $c - h$ is smaller than every later value, so $A_c = c - h$. For odd $c \geq h$ every later gap carries a smaller value, so $A_c = (3h-1-c) - (2h-1-c) = h$. Hence $Q_c - A_c = 0, 1, 1, 0$ in the four cases, which is exactly $\mathbf{1}\{\tau_c = \downarrow\}$. □

The algorithm. Let $\ell \geq 1$ and $h = \ell - 1$. For a training collection T define the fitted function

$$\mathcal{A}(T)(u) = u - G\big(p_T(u), u\big), \tag{5}$$

where

$$G(p, u) = \begin{cases} 3 + \gamma_p + \frac{1}{2}\left(u\,\mathbf{1}\{\tau_p = \uparrow\} + (1-u)\,\mathbf{1}\{\tau_p = \downarrow\}\right), & p \leq 2h-1, \\ 2 + \frac{1}{2}u, & p = 2h, \\ 1 - \frac{p}{2n}, & p > 2h. \end{cases}$$

$\mathcal{A}$ is symmetric and uses only the covariates. With $Y = X \sim \mathrm{Unif}[0,1]$, the residual of a point (u,u) under $\mathcal{A}(T)$ is exactly $G(p_T(u),u) > 0$, and the test residual $Y_{n+1} - \widehat{\mu}_{-i}(X_{n+1})$ is positive. For $\ell = 1$ the first case of G is void.

Proof of the lower bound in Theorem 2. Assume $\ell \geq 1$; since $\alpha \leq 1/2$ we have $2\ell - 1 \leq n$. Fix a realization with distinct values and ranks as above; call ranks $1, \ldots, 2\ell - 1$ *special*. Compare ranks $a < b$ with the fit that omits both; by (3) the positions are $a-1$ and $b-2$. (i) If both are special, then $a-1, b-2 \leq 2h-1$. The levels $3+\gamma_p$ are separated by 1 and the tie-break term lies in $[0, 1/2]$, so the comparison is exactly the tournament of Lemma 3. (ii) If a is special and b is not, then $b \geq 2h+2$, so b has position $b-2 \geq 2h$ and residual at most $5/2$. If $a \leq 2h$, the residual of a is at least 3. If $a = 2h+1$, then a has position $2h$ and residual $2 + z_a/2$, while b has either position $2h$ and residual $2 + z_b/2 < 2 + z_a/2$, or a position above $2h$ and residual below 1. In every case a has the larger residual. (iii) Every non-special point loses to all $2\ell - 1$ special points, so it has at most $n - (2\ell - 1) < t$ wins. By Lemma 3 each special point beats exactly $h = \ell - 1$ special points and all $n + 2 - 2\ell$ others, so it has exactly $t = n + 1 - \ell$ wins. Thus the strange set is precisely the set of special ranks. For the test point, $R_i = G(p_{T_i}(X_i), X_i)$ and $\widehat{\mu}_{-i}(X_{n+1}) + R_i < Y_{n+1}$ is equivalent to $R_i < G(p_{T_i}(X_{n+1}), X_{n+1})$, i.e. to $W_{n+1,i} = 1$, where T_i is the training set without point i. So a special test point has t such indices and lies above $\widehat{q}^{+}_{n,\alpha}$. The test point is special with probability $(2\ell - 1)/(n+1)$. □

Remark 4 (Why a construction is needed, and why this one)**.** Under exchangeable (permutation) distributions one can place a regular tournament directly, and Vovk [18] built examples of this kind for leave-one-out cross-conformal prediction. For i.i.d. data and the jackknife+ interval, every comparison must come from a fit that has not seen the two compared points, which rules out most tournaments. The zigzag is not an arbitrary choice: an exhaustive search over all rules of the form "compare positions by a permutation γ, break equal positions by a direction τ" shows that for every $\ell \leq 12$ exactly two rules yield a regular tournament, the zigzag and its mirror image $\gamma \mapsto 2h - 1 - \gamma$ (Section 7).

Proposition 5. *Let $\ell \geq 3$, and consider any rule of the following form: a comparison between points at distinct positions is decided by a fixed strict total order $\succ$ on positions (the point whose position is higher in $\succ$ has the larger residual), and comparisons at equal positions are decided arbitrarily. No such rule realizes the rotational tournament on the special ranks, in which rank a beats rank $b > a$ if and only if $b - a \leq \ell - 1$.*

The zigzag rule is of this form: $c \succ c'$ if and only if $\gamma_c > \gamma_{c'}$ for positions $c, c' \leq 2h-1$, followed by position $2h$ and then by the remaining positions in decreasing order.

Proof. By (3), ranks a and $a+1$ are compared at equal positions, and ranks $a < b$ with $b \geq a+2$ at the distinct positions $c = a-1 < c' = b-2$. For $0 \leq c < c' \leq 2\ell - 3$ the rotational rule therefore requires $c \succ c'$ whenever $1 \leq c' - c \leq \ell - 2$, and $c' \succ c$ whenever $c' - c \geq \ell - 1$. The first condition applied to consecutive positions gives $0 \succ 1 \succ \cdots \succ \ell - 1$, hence $0 \succ \ell - 1$ by transitivity, while the second with $(c, c') = (0, \ell - 1)$ gives $\ell - 1 \succ 0$, a contradiction. □

4 Finite-sample bounds for K-fold CV+

Let $n = Km$. Two elementary facts about the loss counts (2) drive the bounds. First, each X_k is *super-uniform* on $\{0, \ldots, m\}$:

$$\mathbb{P}(X_k \leq j) \leq \frac{j+1}{m+1}, \qquad j = 0, \ldots, m, \tag{6}$$

because, conditionally on the data outside $S_k \cup \{n+1\}$, the $m+1$ residuals of $S_k \cup \{n+1\}$ under $\widehat{\mu}_{-S_k}$ are i.i.d., and among $m+1$ exchangeable values at most $j+1$ can have at most j others at least as large. Second, the tournament argument applies to the augmented fold structure of Barber et al. [2].

Theorem 6. *Let $n = Km$, $\alpha \in (0, 1/2]$, $\ell = \lfloor \alpha(n+1) \rfloor \geq 1$ and $L = \ell - 1$. For every distribution and every symmetric algorithm,*

$$\mathbb{P}\Big(Y_{n+1} \notin \widehat{C}^{\mathrm{CC}}_{n,K,\alpha}(X_{n+1})\Big) \leq \min\{B_{\mathrm{avg}}, B_{\mathrm{cnt}}\},$$

where

$$B_{\mathrm{avg}} = \min_{c \in \{1,\dots,m+1\},\, Kc > L} \frac{K\, c(c+1)}{2(m+1)(Kc - L)},$$
$$B_{\mathrm{cnt}} = \frac{s^*}{n+m}, \qquad s^* = \max\Big\{ s \leq (K+1)m : \ \tbinom{s}{2} - \pi_m(s) \leq s(\ell - 1) \Big\},$$

and $\pi_m(s) = q\binom{m}{2} + \binom{r}{2}$ for $s = qm + r$, $0 \leq r < m$. Moreover $B_{\mathrm{avg}} \leq 2\alpha + \frac{2(1-\alpha)(1-1/K)}{n/K+1}$ and $B_{\mathrm{cnt}} \leq 2\alpha + \frac{1-K/n}{K+1}$, so the bound improves both terms of (1), the first even in the sharper form of Gasparin and Ramdas [7].

Proof. Averaging bound. For an integer $c \geq 1$ and an integer $x \geq 0$, $(c-x)_+ = \sum_{j=0}^{c-1} \mathbf{1}\{x \leq j\}$, so (6) gives $\mathbb{E}(c - X_k)_+ \leq \sum_{j<c} \frac{j+1}{m+1} = \frac{c(c+1)}{2(m+1)}$ for $c \leq m+1$. On the event $\sum_k X_k \leq L$ we have $\sum_k (c - X_k)_+ \geq Kc - L > 0$. Markov's inequality and Lemma 1 give the bound.

Counting bound. Add $m-1$ further i.i.d. test points so that the test points form a fold S_{K+1} of size m. For i, j in different folds let $R_{ij} = |Y_i - \widetilde{\mu}_{-(k(i),k(j))}(X_i)|$, where $\widetilde{\mu}_{-(k,k')}$ is fitted on all points outside folds k and k', and $W_{ij} = \mathbf{1}\{R_{ij} > R_{ji}\}$. Then $\widetilde{\mu}_{-(k,K+1)} = \widehat{\mu}_{-S_k}$, so by Lemma 1 miscoverage implies that the test point beats at least t of its n opponents. Call a point strange if it beats at least t of the n points outside its fold. A strange point has at most $\ell - 1$ non-wins, and two strange points in different folds produce at least one non-win. With s_f strange points in fold f,

$$\tbinom{s}{2} - \textstyle\sum_f \binom{s_f}{2} \leq s(\ell - 1).$$

Since $s_f \leq m$ and $x \mapsto \binom{x}{2}$ is convex, $\sum_f \binom{s_f}{2} \leq \pi_m(s)$, so $s \leq s^*$. Permutations within folds and of whole folds preserve the joint law and act transitively on the $n+m$ points, so $\mathbb{P}(\text{test point strange}) = \mathbb{E}s/(n+m) \leq s^*/(n+m)$.

Comparison with (1). For real $c = k + \theta$ with $\theta \in [0, 1)$ the exact value is $\mathbb{E}(c - U)_+ = \frac{c(c+1)+\theta(1-\theta)}{2(m+1)}$ for U uniform on $\{0, \dots, m\}$, and it dominates $\mathbb{E}(c - X_k)_+$ because $(c-x)_+$ is decreasing and X_k is stochastically larger than U. On each interval $[k, k+1]$ the ratio of this piecewise-linear function to $Kc - L$ is monotone, so the minimum over real c is attained at an integer. Evaluating at $c^\star = (2L + K)/K \leq m + 1$ (valid because $\alpha \leq 1/2$) gives

$$B_{\mathrm{avg}} \leq \frac{2L+K}{K(m+1)} + \frac{K}{8(m+1)(L+K)} \leq \frac{2(L+K)}{K(m+1)} = \frac{2(\ell + K - 1)}{n+K} \leq 2\alpha + \frac{2(1-\alpha)(K-1)}{n+K},$$

using $\ell \leq \alpha(n+1)$, so that $\frac{2\ell}{n+K} \leq 2\alpha - \frac{2\alpha(K-1)}{n+K}$. For the counting bound, $\pi_m(s) = q\binom{m}{2} + \binom{r}{2} \leq \frac{s(m-1)}{2}$, so s^* satisfies $\binom{s^*}{2} - \frac{s^*(m-1)}{2} \leq s^*(\ell - 1) \leq s^*(\alpha(n+1) - 1)$, i.e. $s^* \leq 2\alpha(n+1) + m - 2$. Hence $B_{\mathrm{cnt}} - 2\alpha \leq \frac{2\alpha(n+1)+m-2}{n+m} - 2\alpha = \frac{(m-1)(1-2\alpha)-1}{n+m} \leq \frac{m-1}{n+m} = \frac{1-K/n}{K+1}$. □

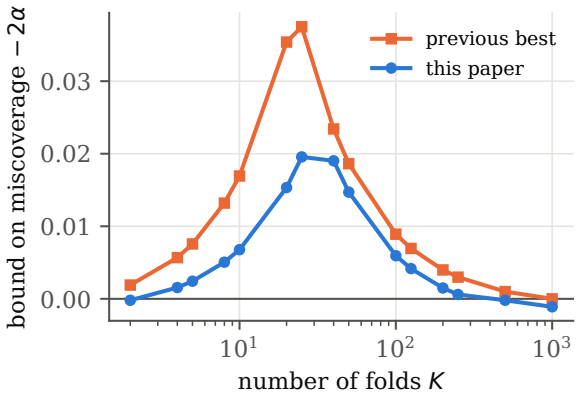


Figure 2: Excess of the worst-case miscoverage bound over 2α for K-fold CV+ and cross-conformal prediction, $n = 1000$, $\alpha = 0.05$. Orange: best previous bound, that is, (1) with the sharper averaging term of Gasparin and Ramdas [7]. Blue: Theorem 6. Negative values mean the clean guarantee $1 - 2\alpha$ holds.

Corollary 7 (The clean $1-2\alpha$ guarantee)**.** *Under the assumptions of Theorem 6: (i) if $K = 2$, then $\mathfrak{W}^{\mathrm{CC}}_{n,2}(\alpha) \le \frac{\ell}{m+1} = \frac{2\ell}{n+2} < 2\alpha$; (ii) if $m = 1$, then $B_{\mathrm{cnt}} = \frac{2\ell-1}{n+1}$; if $m = 2$, then $B_{\mathrm{cnt}} = \frac{2\ell}{n+2} < 2\alpha$. Hence CV+ and cross-conformal prediction cover with probability at least $1 - 2\alpha$, with no extra term, whenever $K = 2$ or the folds have size at most 2.*

Proof. (i) Take $c = \ell \le m + 1$ in B_{avg}: $\frac{2\ell(\ell+1)}{2(m+1)(2\ell-\ell+1)} = \frac{\ell}{m+1}$. (ii) Note $2\ell \le n + 1$ because $\alpha \le 1/2$. For $m = 1$, $\pi_1 \equiv 0$, and $\binom{s}{2} \le s(\ell - 1)$ holds exactly for $s \le 2\ell - 1$, so $s^* = 2\ell - 1$. For $m = 2$, $\pi_2(s) = \lfloor s/2 \rfloor$. The value $s = 2\ell \le (K + 1)m$ satisfies the condition with equality, $\binom{2\ell}{2} - \ell = 2\ell(\ell - 1)$, while every $s \ge 2\ell + 1$ violates it, because $\binom{s}{2} - \lfloor s/2 \rfloor \ge \frac{s(s-2)}{2} > s(\ell - 1)$. So $s^* = 2\ell$. Finally $2\ell/(n + 2) < 2\alpha$ because $\ell \le \alpha(n + 1)$. □

Table 1 and Fig. 2 compare Theorem 6 with the best previous bound. The averaging bound is better for few large folds and the counting bound for many small folds. Their minimum never exceeds the best previous bound, and it was strictly smaller in all 515 cases we checked ($n \le 2000$, $\alpha \le 1/2$, every divisor $K \ge 2$ of n).

5 Two folds: the 3/2 law

5.1 Averaging conditionally i.i.d. p-values

Lemma 8. *Let $V_1, V_2 \ge 0$ be conditionally i.i.d. given a random element Θ, and suppose $\mathbb{P}(V_1 \le u) \le u$ for all $u \in [0, 1]$. Then for every $c \in [0, 1]$,*

$$\mathbb{P}(V_1 + V_2 \le c) \le \tfrac{3}{4}\, c.$$

The constant is sharp: for every $c \in (0, 1]$ there is such a pair with $V_1 \sim \mathrm{Unif}[0, 1]$ exactly and $\mathbb{P}(V_1 + V_2 \le c) = \frac{3}{4}c$.

Proof. Let $F(u) = \mathbb{P}(V_1 \le u \mid \Theta)$, $a = F(c/2)$ and $d = F(c)$, so $0 \le a \le d \le 1$. If $V_1 + V_2 \le c$, then $\min(V_1, V_2) \le c/2$ and $\max(V_1, V_2) \le c$, hence

$$\mathbb{P}(V_1 + V_2 \le c \mid \Theta) \le d^2 - (d - a)^2 = ad + a(d - a) \le a + \tfrac{d^2}{4} \le a + \tfrac{d}{4}.$$

n	K	m	2α	previous	B_{avg}	B_{cnt}	ours
1000	2	500	0.1000	0.1019	0.0998	0.3693	**0.0998**
1000	5	200	0.1000	0.1076	0.1024	0.2200	**0.1024**
1000	10	100	0.1000	0.1169	0.1068	0.1782	**0.1068**
1000	20	50	0.1000	0.1354	0.1153	0.1400	**0.1153**
1000	25	40	0.1000	0.1375	0.1196	0.1298	**0.1196**
1000	50	20	0.1000	0.1186	0.1401	0.1147	**0.1147**
1000	100	10	0.1000	0.1089	0.1783	0.1059	**0.1059**
1000	500	2	0.1000	0.1010	0.3695	0.0998	**0.0998**
1000	1000	1	0.1000	0.1000	0.5258	0.0989	**0.0989**
100	10	10	0.2000	0.2818	0.2479	0.2455	**0.2455**
100	20	5	0.2000	0.2381	0.3030	0.2095	**0.2095**

Table 1: Worst-case miscoverage bounds for K-fold CV+ and cross-conformal prediction ($\alpha = 0.05$ for $n = 1000$, $\alpha = 0.1$ for $n = 100$). "Previous" is the best previous bound: the minimum of the counting term of (1) and the sharper averaging term of Gasparin and Ramdas [7]. Rows with $K = n$ are the jackknife+, where B_{cnt} is exact (Theorem 2).

Taking expectations, $\mathbb{P}(V_1 + V_2 \le c) \le \mathbb{P}(V_1 \le c/2) + \frac{1}{4}\mathbb{P}(V_1 \le c) \le \frac{c}{2} + \frac{c}{4}$. For sharpness, with probability c let Θ be "special" with $A \sim \text{Unif}[0, c/2)$, and given A let V_1, V_2 be i.i.d. uniform on $\{A, c - A\}$; otherwise let V_1, V_2 be i.i.d. $\text{Unif}(c, 1]$. Then $V_1 \sim \text{Unif}[0, 1]$. In the special case the three coin combinations (A, A), $(A, c - A)$, $(c - A, A)$ have sum at most c and the fourth does not; otherwise the sum exceeds c. So $\mathbb{P}(V_1 + V_2 \le c) = \frac{3}{4}c$. □

Under arbitrary dependence the corresponding constant is 1 ($\mathbb{P}(V_1 + V_2 \le c) \le c$, attained by antithetic pairs), which is the source of the factor 2 in Vovk and Wang [19]. Exchangeability alone does not help [5]; the antithetic example is exchangeable but is not a mixture of i.i.d. laws, and Lemma 8 shows no such mixture can attain it.

Corollary 9 (Two random splits)**.** *Let p_1, p_2 be split conformal p-values for the test point computed from two random splits of the same data, with any fixed split sizes and any algorithm. Assume that the two splits are drawn independently of each other and of the data, uniformly and with replacement (so they may coincide), and that the algorithm is deterministic or uses fresh independent randomness for each split. Then for $\alpha' \in (0, 1/2]$,*

$$\mathbb{P}\Big(\tfrac{p_1(Y_{n+1}) + p_2(Y_{n+1})}{2} \le \alpha'\Big) \le \tfrac{3}{2}\,\alpha'.$$

So averaging the two p-values and thresholding at $2\alpha/3$ guarantees coverage $1 - \alpha$ in finite samples, instead of the threshold $\alpha/2$ implied by the general factor 2.

Proof. Given the data $(Z_1, \ldots, Z_{n+1})$ the two splits, together with their internal randomness, are i.i.d., so $p_1(Y_{n+1}), p_2(Y_{n+1})$ are conditionally i.i.d.; each is super-uniform by the validity of split conformal prediction. Apply Lemma 8 with $c = 2\alpha'$. □

Independence of the two draws matters: if the second split is drawn from the splits not yet used, the two p-values are exchangeable but not conditionally i.i.d., and Lemma 8 does not apply.

5.2 Two-fold CV+ and cross-conformal prediction

With $K = 2$ the model used for fold 1 is trained on fold 2 and vice versa. For a fold F used as training data define the score $s_F(z) = |y - \mathcal{A}(F)(x)|$ and its tail probability $v_F(z) = \mathbb{P}_{Z' \sim P}\big(s_F(Z') \ge s_F(z)\big)$.

Theorem 10. *Let $K = 2$, $n = 2m$, $\alpha \in (0, 1/2)$ and $L = \lfloor \alpha(2m+1) \rfloor - 1$.*

(i) Upper bound. *For every distribution, every algorithm (symmetric or not) and every $\varepsilon > 0$ with $L/m + 2\varepsilon \le 1$,*

$$\mathbb{P}\Big(Y_{n+1} \notin \widehat{C}^{\mathrm{CC}}_{n,2,\alpha}(X_{n+1})\Big) \le \tfrac{3}{4}\big(L/m + 2\varepsilon\big) + 2e^{-2m\varepsilon^2}.$$

In particular $\mathfrak{W}^{\mathrm{CC}}_{2m,2}(\alpha) \le \frac{3}{2}\alpha + \frac{3}{2}\sqrt{\log m/(2m)} + \frac{2}{m}$ once $2\alpha + \sqrt{2\log m/m} \le 1$.

(ii) Lower bound. *There is a distribution and a symmetric algorithm with*

$$\mathbb{P}\Big(Y_{n+1} \notin \widehat{C}^{\mathrm{CV+}}_{n,2,\alpha}(X_{n+1})\Big) \;\ge\; \tfrac{3}{2}\alpha - \tfrac{3}{4}\sqrt{\log m/m} - \tfrac{3}{2m}.$$

(iii) Consequently $\lim_{m\to\infty} \mathfrak{W}^{\mathrm{CV+}}_{2m,2}(\alpha) = \lim_{m\to\infty} \mathfrak{W}^{\mathrm{CC}}_{2m,2}(\alpha) = \frac{3}{2}\alpha$.

Proof of (i). Write $Z = Z_{n+1}$, $V_1 = v_{S_1}(Z)$ and $V_2 = v_{S_2}(Z)$. In (2), $X_1 = \#\{i \in S_1 : s_{S_2}(Z_i) \ge s_{S_2}(Z)\}$ and $X_2 = \#\{i \in S_2 : s_{S_1}(Z_i) \ge s_{S_1}(Z)\}$. Conditionally on (S_2, Z) the points of S_1 are i.i.d. and independent of the fitted model, so $X_1 \sim \mathrm{Bin}(m, V_2)$; Hoeffding's inequality [10] gives $\mathbb{P}(X_1 < m(V_2 - \varepsilon)) \le e^{-2m\varepsilon^2}$, and likewise for X_2. Outside these two events, $X_1 + X_2 \le L$ forces $V_1 + V_2 \le L/m + 2\varepsilon$. Conditionally on Z the folds S_1, S_2 are i.i.d., so V_1, V_2 are conditionally i.i.d.; and for each fixed F the variable $v_F(Z)$ is super-uniform, hence so is V_1. Lemmas 1 and 8 give the bound. For the second statement take $\varepsilon = \sqrt{\log m/(2m)}$ and use $L/m < 2\alpha$. □

Proof of (ii): the construction. Let $U, W \sim \mathrm{Unif}[0,1]$ be independent, $X = (U, W)$ and $Y = U$. Fix $\beta \in (0, \alpha)$ and define two measure-preserving bijections of $[0,1)$ (up to null sets) that differ only on $[0, 2\beta)$:

$$V_0(u) = \begin{cases} u, & u < \beta, \\ 3\beta - u, & \beta \le u < 2\beta, \end{cases} \qquad V_1(u) = \begin{cases} 2\beta - u, & u < \beta, \\ u - \beta, & \beta \le u < 2\beta, \end{cases}$$

and $V_0(u) = V_1(u) = u$ for $u \ge 2\beta$. The algorithm computes a fair coin from its training fold, $\kappa(F) = \mathbf{1}\{\mathrm{frac}(\sum_{i\in F} W_i) < 1/2\}$, and returns $\mathcal{A}(F)(u, w) = u - \big(1 - V_{\kappa(F)}(u)\big)$. It is symmetric. The residual of a point (u, w, u) is $1 - V_{\kappa(F)}(u)$, so $v_F(u) = V_{\kappa(F)}(u)$; the test residual is positive, and by Lemma 1 miscoverage is exactly the event $X_1 + X_2 \le L$. The coins of the two folds are independent fair coins, independent of all U's. Given the test value u and the coins, $X_1 \sim \mathrm{Bin}(m, V_{\kappa(S_2)}(u))$ and $X_2 \sim \mathrm{Bin}(m, V_{\kappa(S_1)}(u))$ are independent.

For $u < 2\beta$ write $a = u \bmod \beta \in [0, \beta)$; the pair $\{V_0(u), V_1(u)\}$ equals $\{a, 2\beta - a\}$. So for three of the four coin outcomes $\mathbb{E}(X_1 + X_2) \le 2\beta m$. Let $\delta := \alpha - \beta$. Since $L + 1 = \ell > \alpha(2m+1) - 1 \ge 2\alpha m - 1$, we have $L + 1 - 2\beta m > 2m\delta - 1$. If $2m\delta \ge 1$, Hoeffding's inequality for the $2m$ independent indicators gives, for these three outcomes,

$$\mathbb{P}(X_1 + X_2 > L) \le \exp\Big(-\tfrac{(L+1-2\beta m)^2}{m}\Big) \le \exp\Big(-\tfrac{(2m\delta - 1)^2}{m}\Big).$$

Therefore the miscoverage is at least $2\beta \cdot \frac{3}{4} \cdot \big(1 - e^{-(2m\delta-1)^2/m}\big)$. Choosing $\delta = \frac{1}{2}\sqrt{\log m/m} + \frac{1}{2m}$, so that $2m\delta - 1 = \sqrt{m \log m} \ge 0$, gives $e^{-(2m\delta-1)^2/m} = 1/m$. Then $\frac{3}{2}(\alpha - \delta)(1 - \frac{1}{m}) \ge \frac{3}{2}\alpha - \frac{3}{4}\sqrt{\log m/m} - \frac{3}{2m}$ because $\alpha < 1/2$, which is the stated bound (it is trivial unless $\delta < \alpha$). □

Remark 11 (Finite m)**.** For every m the best proved bound is $\min\{\ell/(m+1), (\mathrm{i})\}$, which lies below 2α (Corollary 7). Convergence is slow on both sides (Fig. 1, right). At $\alpha = 0.1$ the construction, optimized over β, reaches 0.1164 at $m = 100$, 0.1443 at $m = 10^4$ and 0.1497 at $m = 10^7$. The bound (i) drops below 2α only from $m \approx 3.3 \times 10^3$ on, and equals 0.180 at $m = 10^4$ and 0.151 at $m = 10^7$. The upper bound loses a term of order $\sqrt{\log m/m}$ in the concentration step, where it must handle the fact that each fold is both a training set for one model and a query set for the other. We do not know the rate at which the true worst case approaches $\frac{3}{2}\alpha$, and pinning down the exact worst case at moderate m is open.

6 Beyond two folds

For $K \geq 3$ the model for fold k is trained on the other $K-1$ folds, so any two models share $K-2$ folds, and the fold p-values need not be conditionally i.i.d. given the test point. An adversary can nevertheless make them exactly conditionally i.i.d., and this gives lower bounds for every K.

Averaging conditionally i.i.d. p-values. For $B \geq 2$ and $u \in (0, 1]$ let

$$g_B(u) = \sup \ \frac{1}{u}\,\mathbb{P}\Big(\frac{1}{B}\sum_{b=1}^{B} p_b \leq u\Big),$$

where the supremum is over all super-uniform $p_1, \ldots, p_B$ that are i.i.d. conditionally on some random element. Four facts frame this constant. (a) g_B is non-increasing. Given a family with ratio r at level u and $\lambda \in (0, 1]$, use the scaled family $(\lambda p_b)_b$ with probability λ and the constant family $p_b \equiv 1$ otherwise. The result is super-uniform and conditionally i.i.d., and it has ratio r at level λu. We write $f_B = \lim_{u\to 0} g_B(u) = \sup_u g_B(u)$. (b) $g_2(u) = 3/2$ for $u \leq 1/2$, by Lemma 8. (c) $g_B \leq 2$, by the arbitrary-dependence bound of Vovk and Wang [19]. (d) $g_B(u) \to 2$ as $B \to \infty$, for every $u < 1/2$. Fix $\delta \in (0, 1)$ and let $q = (2-2\delta)u < 1$. With probability q let the p_b be i.i.d. uniform on $[0, q]$, and otherwise i.i.d. uniform on $(q, 1]$. Each p_b is then uniform, and $\mathbb{P}(\bar{p} \leq u) \geq q\,\mathbb{P}\big(\bar{U}_B \leq \frac{1}{2-2\delta}\big) \to q$, where $\bar{U}_B$ is the mean of B i.i.d. uniforms. So $\liminf_B g_B(u) \geq 2 - 2\delta$. Fact (d) is the finite-B counterpart of the sharpness of the factor 2 for conditional expectations of p-values [17, 22], which are the limits of the mean as $B \to \infty$.

Lower bounds for K-fold CV+.

Proposition 12. *For every $K \geq 2$ and $\alpha \in (0, 1)$,*

$$\liminf_{m\to\infty} \mathfrak{W}^{\mathrm{CV}+}_{Km,K}(\alpha) \ \geq \ \alpha\, g_K(\alpha).$$

Proof. Fix $\alpha' \in (0, \alpha)$ and super-uniform $p_1, \ldots, p_K$ that are i.i.d. given a random element Θ.

Step 1: uniform marginals. Let F be the distribution function of p_1, and let $\xi_1, \ldots, \xi_K$ be i.i.d. uniform and independent of everything else. The randomized probability integral transforms $p'_k = F(p_k-) + \xi_k\big(F(p_k) - F(p_k-)\big)$ are uniform, satisfy $p'_k \leq F(p_k) \leq p_k$, and are i.i.d. given Θ. We may take Θ to be the conditional law of p'_1, a random element of a Polish space, and write F_Θ for its distribution function.

Step 2: a representation. Split the binary digits of a uniform variable U into two independent uniform variables. Use the first to generate Θ_U with the law of Θ, and take the second as S_U. Define $\varphi(h, u) = F^{-1}_{\Theta_u}\big(\mathrm{frac}(h + S_u)\big)$ for $h \in [0, 1)$, with F^{-1} the quantile function. For fixed h, $\mathrm{frac}(h+S_U)$ is uniform and independent of Θ_U, so $\varphi(h, U)$ is uniform. If $h_1, \ldots, h_K$ are i.i.d. uniform

and independent of U, then given U the values $\varphi(h_k, U)$ are i.i.d. with distribution function F_{Θ_U}. So $(\varphi(h_k, U))_k$ has the law of $(p'_k)_k$.

Step 3: the algorithm. Let $X = (U, W)$ with independent uniform coordinates, and $Y = U$. For a training collection T let $h_T = \mathrm{frac}\big(\sum_{(u,w,y)\in T} w\big)$ and $\mathcal{A}(T)(u, w) = u - 1 + \varphi(h_T, u)$. The algorithm is symmetric, and the residual of a point (u, w, u) under $\mathcal{A}(T)$ is $1 - \varphi(h_T, u) \in (0, 1]$. Let h_k be the hash of the training set of the model for fold k. Then $X_k = \#\{i \in S_k : \varphi(h_k, U_i) \le \varphi(h_k, U_{n+1})\}$, and all test residuals are positive. So whenever $\sum_k X_k \le L$, at least $n - L = t$ of the values $\widehat{\mu}_{-S_{k(i)}}(X_{n+1}) + R_i^{\mathrm{CV}}$ lie below Y_{n+1}, and CV+ miscovers.

Step 4: the hashes are exactly i.i.d. Let $T_j = \mathrm{frac}\big(\sum_{i\in S_j} W_i\big)$. These are i.i.d. uniform on the circle $\mathbb{R}/\mathbb{Z}$, and $h_k = \mathrm{frac}\big(\sum_{j\neq k} T_j\big)$. The integer matrix $J - I$, with J the all-ones matrix, has determinant $(-1)^{K-1}(K-1) \neq 0$. So $(T_j)_j \mapsto (h_k)_k$ is a surjective continuous endomorphism of the torus $(\mathbb{R}/\mathbb{Z})^K$, and it maps Haar measure to Haar measure. Hence $h_1, \ldots, h_K$ are i.i.d. uniform for every m, and they are independent of all U_i. Given $U_{n+1} = u$ and the hashes, the counts $X_k \sim \mathrm{Bin}(m, \varphi(h_k, u))$ are independent, because they involve disjoint sets of the U_i.

Step 5: the limit. Here $L = \lfloor \alpha(Km + 1) \rfloor - 1 \ge Km\alpha - 2$. On the event $\sum_k \varphi(h_k, u) \le K\alpha'$, Hoeffding's inequality gives $\mathbb{P}\big(\sum_k X_k > L \mid u, h\big) \le \exp\big(-2(Km(\alpha - \alpha') - 1)^2/(Km)\big)$ once $Km(\alpha - \alpha') \ge 1$, and this tends to 0. By Steps 2 and 3,

$$\liminf_{m\to\infty} \mathbb{P}\big(Y_{n+1} \notin \widehat{C}^{\mathrm{CV+}}_{Km,K,\alpha}(X_{n+1})\big) \;\ge\; \mathbb{P}\Big(\frac{1}{K}\sum_k \varphi(h_k, U) \le \alpha'\Big) = \mathbb{P}(\bar{p}' \le \alpha') \;\ge\; \mathbb{P}(\bar{p} \le \alpha').$$

Taking the supremum over families gives $\liminf_m \mathfrak{W}^{\mathrm{CV+}}_{Km,K}(\alpha) \ge \alpha' g_K(\alpha') \ge \alpha' g_K(\alpha)$, by fact (a). Now let $\alpha' \uparrow \alpha$. □

Corollary 13 (Many folds)**.** *For every $\alpha \in (0, 1/2)$,*

$$\lim_{K\to\infty} \liminf_{m\to\infty} \frac{\mathfrak{W}^{\mathrm{CV+}}_{Km,K}(\alpha)}{\alpha} \;=\; \lim_{K\to\infty} \limsup_{m\to\infty} \frac{\mathfrak{W}^{\mathrm{CC}}_{Km,K}(\alpha)}{\alpha} \;=\; 2.$$

Proof. The lower bound follows from Proposition 12 and fact (d). For the upper bound, Theorem 6 gives $\mathfrak{W}^{\mathrm{CC}}_{Km,K}(\alpha) \le B_{\mathrm{avg}} \le 2\alpha + \frac{2(K-1)}{Km+K}$, which tends to 2α as $m \to \infty$; and $\mathfrak{W}^{\mathrm{CV+}} \le \mathfrak{W}^{\mathrm{CC}}$. □

So the 3/2 law is special to few folds. With two folds the fold p-values are forced to be asymptotically conditionally i.i.d., which caps the damage at $\frac{3}{2}\alpha$; with many folds, conditionally i.i.d. fold p-values already reach the factor 2. For fixed $K \ge 3$ the large-fold worst case lies between $\alpha\, g_K(\alpha)$ and 2α. Because the models share folds, dependent constructions may exceed $\alpha\, g_K(\alpha)$, and whether the limit is strictly below 2α for every fixed $K \ge 3$ is open.

Numerical lower bounds on g_B. For $B \ge 3$ the value of g_B is unknown. To see how large the lower bound of Proposition 12 is for moderate K, we computed lower bounds on g_B by linear programming over a restricted class of families, namely mixtures of two-point laws on a grid (details in Section 7). Each optimal solution is an explicit conditionally i.i.d. super-uniform family, so each entry of Table 2 is a certified lower bound on $g_B(u)$, not a value of g_B. The families put mass only on $[0, \min(B, 3)\, u] \cup \{1\}$, so each bound holds for all $u \le 1/\min(B, 3)$. We solved the program for every $2 \le B \le 30$, and the smallest bound is 1.427, at $B = 3$. With Proposition 12 and fact (b), this gives $\liminf_{m\to\infty} \mathfrak{W}^{\mathrm{CV+}}_{Km,K}(\alpha) \ge 1.427\,\alpha$ for $2 \le K \le 30$ and $\alpha \le 1/3$. Because the families are restricted, the non-monotonicity of the entries in B says nothing about g_B itself. In the other direction, relaxing the event $\{\sum_b p_b \le Bu\}$ to constraints on order statistics and evaluating the relaxation on a grid suggests $f_3 \lesssim 1.57$ and $f_4 \lesssim 1.74$. Turning these into upper bounds requires certifying a polynomial inequality, which we leave open.

B	2	3	4	5	6	8	10	15	20	25	30
$g_B(u) \geq$	1.487	1.427	1.470	1.519	1.505	1.537	1.566	1.603	1.628	1.649	1.668

Table 2: Numerical lower bounds on the averaging constant $g_B(u)$ of B conditionally i.i.d. p-values, each certified by an explicit family from a restricted linear program. They hold for $u \leq 1/\min(B, 3)$ and are rounded down. They are lower bounds only: g_B is unknown for $B \geq 3$. For $B = 2$ the true value is $3/2$ (Lemma 8), and the entry shows the loss from discretization.

Small folds. At the other extreme, the jackknife+ attains the factor 2 exactly (Theorem 2). For folds of fixed size $m \geq 2$ the zigzag construction still works whenever the fold-mates of the compared points are not among the special points, which suggests that the factor 2 is approached as $\alpha m \to 0$. A precise statement is left open.

7 Computational verification

The results of the previous sections are proved. This section records independent computational checks of the constructions and bounds, together with the linear programs behind the numerical lower bounds of Table 2. The scripts are available as supplementary material.

- **Zigzag lemma.** For every $h \leq 100$ we built the tournament of Lemma 3 from its definition and counted all wins directly: it is regular in every case. An exhaustive backtracking search over all pairs (γ, τ) with the defining condition $Q_c - A_c \in \{0, 1\}$ found exactly two solutions for each $\ell = 2, \ldots, 12$, the zigzag and its mirror image.
- **Jackknife+ construction.** We computed all $\binom{n+1}{2}$ leave-two-out comparisons with the algorithm (5): for 54 pairs (n, α) with $2 \leq n \leq 33$ and $0.08 \leq \alpha \leq 1/2$ (five random samples each; the range includes cases with $\ell = 1$ and cases with $2\ell - 1 = n$), and for $(n, \alpha) \in \{(20, 0.15), (40, 0.1), (30, 0.2), (15, 0.3)\}$ (30 samples each). In every case the strange set was exactly the top $2\ell - 1$ ranks, as the proof asserts. A direct simulation of the jackknife+ interval with 2×10^5 trials per setting gave miscoverage 0.2397, 0.1706, 0.3540 for $(n, \alpha) = (20, 0.15), (40, 0.1), (30, 0.2)$, against the exact values 0.2381, 0.1707, 0.3548 ($|z| \leq 1.7$).
- **Two-fold construction.** Averaging over the coins, the miscoverage of the construction in Theorem 10(ii) equals

$$\Phi(1) - \tfrac{1}{2}\Phi(2\beta) + \int_0^\beta \mathbb{P}\big(\mathrm{Bin}(m, a) + \mathrm{Bin}(m, 2\beta - a) \leq L\big)\, da, \qquad \Phi(x) = \int_0^x \mathbb{P}\big(\mathrm{Bin}(2m, u) \leq L\big)\, du,$$

  where Φ has a closed form in regularized incomplete beta functions and the last integrand is a polynomial of degree $2m$ in a, integrated by Gauss–Legendre quadrature (exact when the number of nodes exceeds m; stable to ten digits beyond). The formula agrees with adaptive quadrature of the original integrand to 10^{-16}. We optimized over β for Fig. 1. A direct simulation of the two-fold CV+ interval at $\alpha = 0.1$ gave 0.1093 for $m = 100$, $\beta = 0.09$ (2×10^5 trials; exact 0.1089, $z = 0.6$) and 0.1224 for $m = 200$, $\beta = 0.08$ (10^5 trials; exact 0.1232, $z = -0.8$).
- **K-fold construction.** For the construction of Proposition 12 with the family used in fact (d), with q optimized and $\alpha = 0.1$, the exact finite-m miscoverage is a K-fold convolution of mixed binomial laws. It matched direct simulation of the K-fold CV+ interval: 0.1228 against 0.1217 for $K = 3$, $m = 300$ (2×10^5 trials, $z = -1.5$); 0.1311 against 0.1310 for $K = 5$, $m = 200$

($z = -0.1$); 0.1404 against 0.1405 for $K = 10$, $m = 100$ (10^5 trials, $z = 0.1$). The simulated hashes were uniform and uncorrelated, as Step 4 of the proof asserts.

- **Averaging constants.** The numerical lower bounds in Table 2 come from a linear program in units of u. Atoms sit at the right endpoints of the cells of a grid of step $1/40$ on $(0, v_{\max}]$, $v_{\max} = \min(B, 3)$. The columns are one-point laws and two-point laws whose lower atom is at most 1 and whose weight on the lower atom lies in $\{0.05, 0.10, \dots, 0.95\}$. The constraints cap the mass in each cell by its length; since atoms sit at right endpoints, this gives super-uniformity at every point. The objective is the exact binomial probability that the mean of B draws is at most 1, and the remaining mass is placed at $p = 1$. Reported values are rounded down.
- **K-fold bounds.** B_{avg} coincides, in every case of Table 1, with the optimum of the linear program that maximizes $\mathbb{P}(\sum_k X_k \le L)$ over all joint laws with uniform marginals on $\{0, \dots, m\}$ (solved by cutting planes with a dynamic-programming separation oracle). So no argument that uses only the marginal distribution of the fold losses can improve it.
- **Small cases.** For $K = 2$, $m = 2$ we solved an integer program over all assignments of a total order to every union of two folds in the augmented sample, which relaxes the constraint that the orders come from fitted models and therefore upper-bounds the true worst case. Its optimum is $22/90 \approx 0.244$ for $\ell = 1$ and $1/2$ for $\ell = 2$, below the bounds $1/3$ and $2/3$ of Theorem 6; so for small samples even these improved bounds are not tight.

8 Discussion

The factor 2 in the guarantees for cross-validation conformal methods is exact for the jackknife+ at every sample size, but it is not a law of nature for cross-validation. With two folds the worst case is $\frac{3}{2}\alpha$ in the large-fold limit: the two fold p-values are asymptotically conditionally i.i.d. given the test point, and averaging two such p-values is less fragile than averaging arbitrary ones. The worst case is governed by how much information the fitted models share, and by how many p-values are averaged. Leave-one-out fits share almost everything and can coordinate an adversarial tournament exactly. Two disjoint folds share nothing and can only randomize independently, which costs the factor $\frac{3}{2}$. Many folds can also randomize independently, and the average of many conditionally i.i.d. p-values is again as fragile as an arbitrary average, so the factor 2 returns (Corollary 13).

Practical consequences. (i) Two-fold CV+ and cross-conformal prediction satisfy the clean guarantee $1 - 2\alpha$ at every sample size, and $1 - \frac{3}{2}\alpha - o(1)$ for large folds. (ii) Averaging the p-values of two independent random splits of split conformal prediction and thresholding at $2\alpha/3$ is valid in finite samples. (iii) For $K \in \{5, 10, 20\}$ our finite-sample bounds shrink the extra term of the best previous bound by a factor between 1.15 and 7.5 over $n \in \{100, 200, 500, 1000, 2000, 5000, 10^4\}$ and $\alpha \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ (in one case, $n = 100$, $\alpha = 0.2$, $K = 20$, they remove it), and by a factor of at least 2.1 when $n \ge 1000$. (iv) With many folds the large-fold guarantee cannot be much better than $1 - 2\alpha$ without assumptions on the algorithm.

Limitations and open problems. Our two-fold result is asymptotic in the fold size, and at moderate m the best upper and lower bounds are far apart. For fixed $K \ge 3$ the large-fold worst case lies between $\alpha\, g_K(\alpha)$ and 2α, and we do not know where. The main open questions are: the exact worst case of two-fold CV+ at finite m; the limiting constant for fixed $K \ge 3$; whether folds of fixed size $m \ge 2$ reach the factor 2; and the exact averaging constants g_B for $B \ge 3$. All worst cases

here are attained by deliberately unstable algorithms, so they complement rather than contradict the near-nominal coverage of stable learners [2, 14].

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